

Quan (Kyle) Zhang

808-312-9336 · quanz@hawaii.edu · [ORCID](#) · [ResearchGate](#) · [LinkedIn](#)

Education

Ph.D., Economics (Quantitative Economics & Econometrics), University of Hawai‘i at Mānoa 2023 – Present
Advisor: Sang-Hyop Lee Expected: 2027

M.S., Quantitative Health Sciences, John A. Burns School of Medicine, UH Mānoa 2024 – 2026

M.S., Industrial & Systems Engineering, Northern Illinois University 2012

Research Interests

Health economics and policy; causal inference and applied econometrics for health and social outcomes; **quantitative analytic methods for longitudinal cohort data with complex survey designs (weighting, stratification, replicate-weights variance estimation)**; population aging, frailty dynamics, and the joint determination of health and labor supply over the life course; disaster health economics and community resilience; quantitative health sciences and clinical research methodology, including cost-effectiveness analysis, psychometric measurement, and biomarker-integrated longitudinal study design; health equity and disparities.

Peer-Reviewed Publications

-
1. Zhang, Q. (2026). Do negative social ties accelerate aging in adults, or does aging erode social ties? *Proceedings of the National Academy of Sciences (PNAS)*, 123(20), e2608036123. [DOI](#)
 2. Zhang, Q. (2026). Keep REMS focused on safety, not as a barrier to generics. *Health Affairs*. [DOI](#)
 3. Zhang, Q. (2026). Enhancing life-course analyses of financial disadvantage and depressive mood through longitudinal design and causal inference. *Journal of Epidemiology*. [DOI](#)
 4. Zhang, Q. (2025). Overcoming bias in a fall prevention program in resource-limited areas. *JAMA*. [DOI](#)
 5. Zhang, Q. (2025). Operationalizing distributional cost-effectiveness in genomic medicine: A US-focused comment on Smith et al. *Value in Health*. [DOI](#)
 6. Zhang, Q. (2025). On “The joint impact of greenspace and air pollution on mortality”: Methodological proposals. *International Journal of Epidemiology*. [DOI](#)
 7. Zhang, Q. (2025). A causal roadmap for Child Opportunity Index 2.0. *Public Health Reports*. [DOI](#)

Papers Under Review

-
1. Zhang, Q. (2026). “Balancing Innovation Incentives and Equitable Access for AI Health Technologies: Policy Levers for IP, Transparency, and Solidarity.” Under review, *npj Health Systems*.
 2. Zhang, Q. (2026). “Caregiving as Health Capital Depreciation: Time Allocation Evidence from IPUMS ATUS.” Under review.
 3. Zhang, Q. (2026). “Enrollment Dynamics and Health Equity Effects of California’s Staggered Medi-Cal Expansion to Undocumented Adults.” Under review, *Health Affairs*.
 4. Zhang, Q. (2026). “Health Shocks and the Permanent Labor-Market Exit of Older Europeans.” Under review, *Journal of Population Economics*.
 5. Zhang, Q. (2026). “Inflammatory, Nutritional, and Metabolic Biomarker Profiles in Early-Onset Colorectal Cancer: A 24-Year Cross-Sectional Analysis of NHANES, 2001–2023.” Under review, *JAMA Oncology*.

Working Papers

-
1. Zhang, Q. (2026). “Chronic versus Acute Health Shocks and the Margins of Labor Supply: Evidence from Frailty Dynamics in an Aging Population.”
 2. Zhang, Q. (2026). “Falls, Missing Markets, and Labor Supply: Evidence from China.”
 3. Zhang, Q. (2026). “Later, Longer, Fewer and the Quantity–Quality Trade-off: Evidence from CHARLS.”

Invited Contributions

-
1. Zhang, Q. (Forthcoming). “Multidimensional Poverty Index (MPI): Measurement, Health Deprivations, and Policy Use.” In R. Baikady (Ed.), *The Palgrave Encyclopedia of Global Poverty Research*. Palgrave Macmillan / Springer Nature.
 2. Zhang, Q. (Forthcoming). “Social Determinants of Health and Multidimensional Poverty.” In R. Baikady (Ed.), *The Palgrave Encyclopedia of Global Poverty Research*. Palgrave Macmillan / Springer Nature.

Conference Presentations

-
1. Zhang, Q. (2026). “Chronic versus Acute Health Shocks and the Margins of Labor Supply: Evidence from Frailty Dynamics in an Aging Population.” Southern Economic Association (SEA) 96th Annual Meeting, Houston, TX, November 21–23, 2026.

Research Experience

- Graduate Research Assistant** — UHERO, University of Hawai'i at Mānoa 2024 – Present
Project 1: Maui Wildfire Exposure Study (MauiWES) 2024 – Present
NIH- and state-funded longitudinal cohort. PI: Ruben Juarez. In collaboration with UHERO, JABSOM, and UH Maui College.
- Analyzed mixed-methods data (biomarker, behavioral, socioeconomic) for 2,000+ participants in an IRB-compliant longitudinal study; generated individual-level risk forecasts informing disaster-recovery policy.
 - Designed quantitative instruments and contributed to participant recruitment protocols for ongoing cohort waves. Contributing to a study published in *JAMA Network Open* (2025).
- Project 2: Roots Reborn Program Evaluation* 2025 – 2026
Post-wildfire economic empowerment intervention.
- Designed and executed full evaluation of an IRB-approved human-subjects intervention: Qualtrics survey design → matched-panel construction → pre-post causal analysis → executive summary for policymakers.
 - Built session-level evaluation framework linking program costs to quality-adjusted participant outcomes, producing a submission-ready manuscript with actionable policy recommendations.
- Project 3: Minority-Owned Small Business Financing Study* 2024 – 2025
City & County of Honolulu OER-funded.
- Built 4-module R causal-inference pipeline (PSM, IPW, Heckman selection, mediation), identifying heterogeneous treatment effects of financing barriers across business subgroups.
- GBD Collaborator** — Global Burden of Disease Study 2023, IHME 2024 – Present
- Invited collaborator at the Institute for Health Metrics and Evaluation, University of Washington. Provided substantive peer review for five GBD 2023 estimation manuscripts: dietary risks, colon/rectum cancer, environmental exposures, type 1 diabetes, and antimicrobial resistance (Western Pacific).
- Research Assistant (Co-Op)** — Argonne National Laboratory, TRACC 2011
- Developed regression models for urban evacuation timing in collaboration with NASA. Applied simulation, Design of Experiment, and statistical modeling for Chicago evacuation planning.

Prior Industry Experience

- Program Manager**, Microsoft Corporation, SQL Server Data Warehousing (APS), Redmond, WA 2014 – 2015
Managed \$4M delivery pipeline of cloud data-warehouse appliances; developed logistics and cost model for Parallel Data Warehousing infrastructure.
- Project Management Specialist**, Quanta Computer (Amazon AWS NPI), Bellevue, WA 2013 – 2014
Coordinated New Product Introduction releases for AWS cloud servers; managed BOM provisioning and cross-cultural supply chain operations between North American and Asian markets.
- Project Engineer (Co-Op)**, Caterpillar Inc., Manufacturing Strategy, Aurora, IL 2012
Delivered NPV-based analysis projecting ~\$1M/year savings across warehouse-relocation scenarios; achieved 60% reduction in urgent-order tardiness via Six Sigma DMAIC.

Teaching Experience

- Instructor**, Introduction to Statistics, UH Mānoa Department of Economics Summer 2025
Teaching Assistant, Principles of Microeconomics & Macroeconomics, UH Mānoa 2023 – 2024
Teaching Assistant, Arena Simulation, Operations Research, Decision Analysis, NIU ISE 2010 – 2012

Awards & Honors

- | | |
|------------|--|
| 2026 | Research Stipend, College of Social Sciences, University of Hawai'i at Mānoa |
| 2026 | AWARD Network Summer Institute, University of California, San Francisco (NIA/NIH-funded) |
| 2026 | Bernie Campbell Fellowship Fund Award, University of Hawai'i at Mānoa |
| 2025 | Burnham O. Campbell Fund Award, University of Hawai'i at Mānoa |
| 2024, 2025 | Hung Family Scholarship Award |
| 2024 | Jordan & Cara Odo Foundation Leadership Scholarship |
| 2023 | Achievement Award, Department of Economics, UH Mānoa |

Professional Service

- Peer Reviewer**, *Public Health in Practice* (Elsevier), 2025–Present
Collaborator & Peer Reviewer, Global Burden of Disease Study 2023, Institute for Health Metrics and Evaluation (IHME), University of Washington, 2024–Present
- Global Dietary Risks, 2026
 - Global Colon & Rectum Cancer (IHME ID: 4219), 2026
 - Global Environmental Exposures (IHME ID: 6176), 2026

- Global Type 1 Diabetes (IHME ID: 5338), 2026
- Antimicrobial Resistance in the Western Pacific Region (IHME ID: WHO AMR), 2026

Professional Memberships

American Economic Association (AEA) · American Society of Health Economists (ASHEcon) · International Health Economics Association (iHEA) · Gerontological Society of America (GSA) · Hawaii Economic Association (HEA) · Pi Gamma Mu

Technical Skills

Statistical Software: R (tidyverse, fixest, modelsummary, survey, srvyr, ggplot2, Shiny), Stata 18 (including `svy` suite for complex-survey estimation), Python (pandas, statsmodels, scikit-learn), SQL, Git.

Complex Survey & Longitudinal Methods: Stratified multistage cluster designs, longitudinal and cross-sectional weighting, replicate-weights variance estimation (BRR, jackknife, Taylor linearization), wave-to-wave attrition adjustment, nonresponse weighting.

Causal Inference & Econometrics: Two-way fixed effects, difference-in-differences, instrumental variables (2SLS), event study designs, regression discontinuity, propensity score matching, inverse probability weighting, Heckman selection models, mediation analysis, sensitivity analysis (Oster 2019; Cinelli–Hazlett 2020), Lee bounds, specification curve analysis, wild cluster bootstrap, A/B testing frameworks, power analysis.

Machine Learning & Computational Methods: Double/debiased machine learning (DML), causal forests, LASSO/elastic net, multiple imputation (MICE).

Quantitative Health Sciences: Frailty index construction (Rockwood deficit accumulation), item response theory (IRT), confirmatory factor analysis (CFA), GAD-7/CES-D psychometric scoring, cost-effectiveness analysis (CEA/DCEA), survey design (Qualtrics).

Survey & Data Platforms: CHARLS, GBD, MEPS, HRS, BRFSS, NHANES, IPUMS (USA/CPS/International), ATUS, SHARE.

Certifications

Google Business Intelligence Professional Certificate · Google Data Analytics Professional Certificate · Certified Six Sigma DMAIC Green Belt (Caterpillar Inc.)

Languages

English (fluent) · Mandarin Chinese (native)